

The exploration cost of boundary and population effects

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Abstract

Adaptive assignment policies that increasingly favor one action, such as softmax policies over reward-model scores, produce logs whose value for causal inference depends on the effect being estimated. We price learning by exploration loss: the cumulative value, measured in the deployer’s own objective, forgone whenever a non-preferred action is logged. In a model with a one-dimensional score, Gaussian outcomes and Lipschitz mean functions, logs from one shrinking temperature yield Wald intervals for the effect at the policy’s decision boundary that are asymptotically valid uniformly over the model, while exploration loss tends to zero. In the same model, no design and estimator can estimate the population average effect uniformly consistently unless worst-case exploration loss diverges. The separation relies on the operational cost vanishing at the boundary; with a fixed cost per deviation, boundary inference also requires divergent loss. Without smoothness restrictions, uniform precision for the population effect requires loss that, as the target error shrinks, is of the order of the squared mean root gap divided by the squared error, independent of sample size. A single common temperature needs loss growing almost linearly in the sample size, whereas uniform mixing keeps it bounded.

Keywords: adaptive experiments, limited overlap, regression discontinuity, minimax lower bounds, generative AI, off-policy evaluation

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1 Introduction

Many deployed systems choose actions with a policy that becomes more decisive as it gains confidence: bandit algorithms, recommender and ranking systems, algorithmic eligibility rules, and conversational AI systems that select a response strategy through a softmax over reward-model scores. Their logs record the action taken and often its probability.

When such a policy is known and increasingly deterministic, which causal effects can the logs deliver, and what must be paid to learn the others?

Conversational AI makes the question concrete. Conversations with large language models change political attitudes [1, 2], and the strategies such systems use, such as presenting evidence or asking a question, are chosen adaptively. Regulation and replication concern the effects of these strategies, and the logs of deployed systems are the evidence an analyst inherits.

The answer depends on the target. We distinguish two:

- the *boundary effect*, the effect among contexts where the policy is indifferent between the two actions; and
- the *population average effect* over all contexts the deployment serves.

The first is the effect for units at the margin of the current policy, the quantity that governs small changes to the policy’s threshold, in the spirit of marginal policy-relevant treatment effects [3]; the second is the average outcome change from assigning one action rather than the other across the whole context population. Estimating one well does not substitute for estimating the other.

We price learning in the deployer’s own currency. Each time the logged action differs from the one the deployer’s objective prefers, the deployer gives up an operational value gap $g(h)$; the cumulative sum is the *exploration loss* R_n . For a system that chooses by a score, the natural cost of an exploratory action is the score difference it forgoes: small where the policy is nearly indifferent, large where it is confident. This cost is defined by the deployer’s objective rather than by the outcome under study, so it can be computed at the design stage.

This paper gives lower bounds stated in exploration loss that hold for every estimator and every, possibly adaptive, design, and pairs them with boundary inference in the same model. The combination shows that, under one sharpening temperature, the boundary effect admits valid inference while cumulative exploration loss tends to zero, whereas the population effect cannot be learned uniformly without divergent loss. Its ingredients are familiar, namely limited overlap, local randomization near a threshold and cost-weighted allocation; the contribution is to place both targets in one model on one cost scale, and to make precise when the separation fails. It depends on the cost vanishing at the boundary: if every deviation carries a fixed cost, boundary inference also requires divergent loss (Proposition 1).

Figure 1 shows the construction behind the population-side lower bound. Two outcome models agree near the boundary and differ only for the unpreferred action away from it; a sharpening policy almost never takes that action there.

Contributions.

1. *Separation in one model* (Theorems 1 and 3). With Gaussian outcomes and globally Lipschitz mean functions, a common temperature $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, gives Wald intervals for the boundary effect that are asymptotically valid uniformly over the model

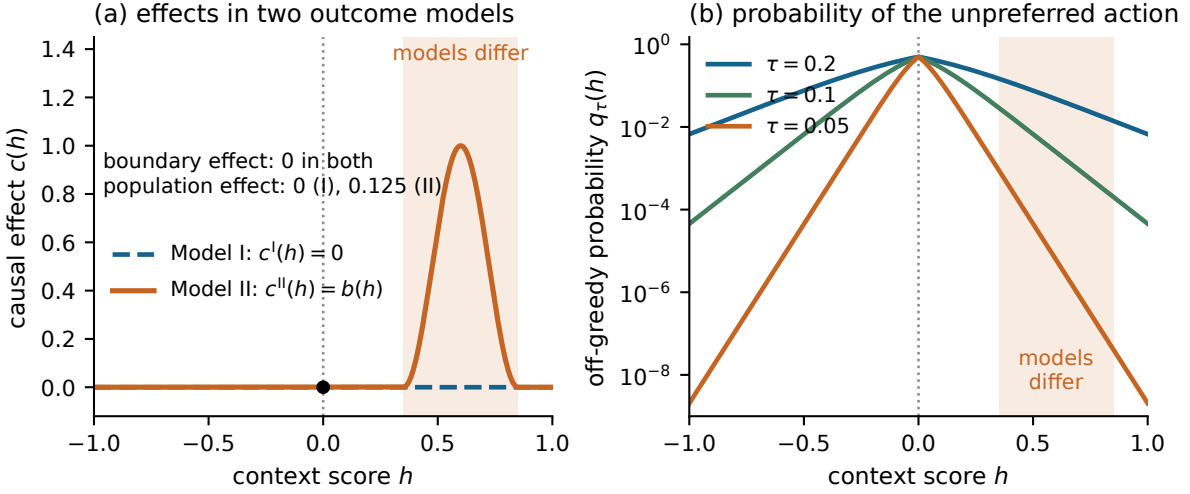


Figure 1: The separation construction, with $H \sim U(-1, 1)$, $g(h) = |h|$, $u_1 = 0.35$, $u_2 = 0.85$ and $t = 1$ (Theorem 3, Model I is μ^0 and Model II is μ^t). Left: the two effect functions have the same boundary effect, zero, but population effects 0 and 0.125; the mean of the preferred action is the same in both models. Right: the probability of the unpreferred action under three temperatures. Shading marks the region in which the models differ. The curves are exact functions of the construction, not estimates.

while $R_n \rightarrow 0$; in the same model, uniformly consistent estimation of the population effect requires $\sup R_n \rightarrow \infty$, and there are two members with equal boundary effects that no estimator distinguishes when $R_n \rightarrow 0$.

2. *An information-exploration inequality* (Theorem 2) for all estimators and adaptive designs in the model of bounded mean functions, attained in order by a gap-based design (Proposition 2).
3. *Consequences for design* (Section 7): one common temperature is costly for the population effect, while uniform mixing, context-dependent temperatures and suitable temperature mixtures are not; and the shape of the operational gap determines whether boundary inference is cheap (Proposition 1).

Scope. All results concern one decision per unit, with independent contexts and a known assignment probability. The score Δ and the preferred action are fixed functions of the context: they are not learned from the outcomes, so “adaptive” refers to sharpening and to designs whose probabilities may depend on past data, not to a bandit that learns which action is better. The boundary results use a one-dimensional score. The lower bounds assume Gaussian outcomes with known variance and hold in models without extrapolation structure: where data near the boundary identify the population effect, as under a constant effect, the separation disappears. Multi-turn conversations, estimated or learned scores and effects of realized text are outside the formal results.

Organization. Section 2 sets out the model and Section 3 states the results informally. Sections 4–6 give boundary inference, the population lower bounds and the separation in one model; Section 7 treats temperature as a design lever; Section 8 reports simulations; Section 9 concludes. Proofs are in the appendix.

Relation to existing work. *Boundary estimands and local randomization.* Lee [4] interprets regression discontinuity as local randomization under imprecise control of the running variable. Abdulkadiroğlu et al. [5, 6] use propensity scores induced by assignment mechanisms, including local propensity scores near tie-breaking cutoffs, and Narita and Yata [7] show that overlap-type weights built from an approximate propensity score of a known algorithm converge to a boundary estimand and require undersmoothing; these are the closest antecedents for Theorem 1. There the smoothing is chosen by the analyst in covariate space; here it is a temperature chosen by the system in score space, and the exploration loss is computed jointly with the inferential guarantee. Eckles et al. [8] use known assignment noise near a cutoff with a fixed noise scale. Overlap weights and their target population are due to Li, Morgan and Zaslavsky [9], and Humphreys [10] shows that the fixed-effects estimand weights stratum effects by the assignment variance $p_j(1 - p_j)$, the same weighting, which under a sharpening policy concentrates on the decision boundary.

Exploration and estimation. Trade-offs between reward and estimation error appear for multi-armed bandits in Erraqabi et al. [11] and Simchi-Levi and Wang [12]. In a non-contextual setting, Liang and Bojinov [13] mix any bandit algorithm with a Bernoulli design whose weight vanishes slowly, keeping anytime-valid inference on the average effect; our recommendation of uniform mixing has the same logic. Inference from adaptively collected data with vanishing propensities is treated by Hadad et al. [14]; adaptive designs for treatment effects and for policy choice by Kato et al. [15] and Kasy and Sautmann [16], with applications in political science reviewed by Offer-Westort, Coppock and Green [17]. Conservative bandits keep cumulative reward above a baseline policy’s [18], and Wan, Kveton and Song [19] optimize exploration for policy evaluation under a value constraint. Tucker and Joachims [20] design logging policies that augment existing logs to reduce off-policy evaluation variance, and Douglas, Persson and Provost [21] derive optimal logging policies for inverse-propensity estimation and report that softmax logging performs worse than top- k logging in simulation. Koppel et al. [22] extend inverse-gap-weighting contextual bandits to balance regret against the quality of average-treatment-effect estimates. The allocation $q \propto g^{-1/2}$ has the form of optimum stratified allocation with costs [23], and Ravichandran et al. [24] derive the analogous cost-weighted rule for factorial experiments. We do not claim the allocation as new; what we add is that the corresponding constant bounds every estimator and adaptive design when loss is paid only for non-preferred actions. Li and Imai [25] compare evaluating an individualized treatment rule by randomizing units to the rule with evaluation from a standard experiment, without a cost for assignments against the rule. Cesa-Bianchi et al. [26] show that Boltzmann exploration with a common schedule can fail for regret; Theorem 4 is an estimation counterpart.

Limited overlap and off-policy evaluation. Lower bounds for off-policy value estimation

[27–29] and limited-overlap asymptotics and trimming [30–34] supply the information-theoretic ingredients; the collapse of inverse weighting under shrinking propensities is irregular identification in the sense of Khan and Tamer [30]. Ring and Schomaker [35] diagnose stochastic positivity problems and use the diagnosis to choose alternative estimands; our results give a formal counterpart for known, sharpening policies, stating which estimand remains learnable and at what cost. Khan, Saveski and Ugander [36] bound effects without overlap under smoothness, the kind of extrapolation structure our impossibility excludes.

Generative treatments. Athey, Imbens and Ji [37] identify effects of content features from stochastic algorithms using logged unexposed candidates and probability ratios; their estimand averages over co-occurring pairs with trimming and does not conflict with our bounds. Fresh and Shin [38] give a taxonomy of conversational estimands, and Imai and Nakamura [39, 40] study effects of realized text features; we work at the level of strategy assignment with the generator held fixed.

2 Setup

Treatments. Units $i = 1, \dots, n$ arrive with contexts $H_i \sim F$. The system chooses a strategy $A_i \in \{0, 1\}$ and produces a message $M_i \sim G(\cdot \mid A_i, H_i)$ from a fixed generation kernel G , which includes token-level sampling settings; the outcome is Y_i . The treatment is the assignment of the strategy with G held fixed, so potential outcomes $Y_i(a)$ include the randomness of the message [41, 42]; changing G changes the treatment. We assume:

- (C1) *consistency*: $Y_i = Y_i(A_i)$;
- (C2) *independent units*: $(H_i, Y_i(0), Y_i(1))$ are independent and identically distributed across i , with no interference;
- (C3) *sequential ignorability with known probabilities*: A_i is independent of $(Y_i(0), Y_i(1))$ given $(H_i, \mathcal{F}_{i-1})$, where $\mathcal{F}_{i-1}$ is the data of units $1, \dots, i-1$, and $\mathbb{P}(A_i = 1 \mid H_i, \mathcal{F}_{i-1})$ is known.

Then the law of Y_i given $(A_i = a, H_i = h, \mathcal{F}_{i-1})$ is that of $Y(a)$ given $H = h$, so statements about conditional outcome laws are statements about potential outcomes.

Deployer. The deployer has a score gap $\Delta(h) = s_1(h) - s_0(h)$ from its own objective, a preferred (greedy) action $a^*(h) = \mathbf{1}\{\Delta(h) > 0\}$, and an operational value gap $g(h) \geq 0$, the value its objective loses by choosing $\bar{a}(h) = 1 - a^*(h)$. We write $g = \phi(|\Delta|)$ when the gap is a function of the score. The score and the preferred action are fixed functions of the context.

Running example. A conversational system chooses between presenting evidence ($A = 1$) and asking a question ($A = 0$). A reward model scores both strategies given the conversation so far, H ; $\Delta(H)$ is the score difference, and the system logs the probability with which it chose evidence. The operational gap $g(H)$ is the loss in the deployer’s

objective, for example predicted engagement, from using the lower-scoring strategy. The outcome Y is the person's attitude after the conversation, measured separately and not used by the policy.

Probabilities and designs. We write

$$e(h) = \mathbb{P}(A = 1 \mid H = h), \quad q(h) = \mathbb{P}(A \neq a^*(h) \mid H = h)$$

for the probability of action 1 and of the non-preferred action. A design is non-adaptive if these probabilities do not depend on past data. A *common temperature* design uses one deterministic $\tau = \tau_n$ and

$$e_\tau(h) = \text{sig}(\Delta(h)/\tau), \quad q_\tau(h) = \text{sig}(-|\Delta(h)|/\tau), \quad \text{sig}(x) = 1/(1 + \exp(-x)).$$

Exploration loss. The expected cumulative operational loss is

$$R_n = \mathbb{E} \left[\sum_{i=1}^n g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\} \right], \quad (1)$$

and $R_n = n\mathbb{E}[g(H)q(H)]$ for non-adaptive designs. Under adaptive designs R_n may depend on the outcome law through past observations.

Estimands. Let $\mu_a(h) = \mathbb{E}[Y(a) \mid H = h]$ and $c = \mu_1 - \mu_0$. The population average effect is $\theta = \mathbb{E}[c(H)]$. With a one-dimensional score $\Delta(h) = h$ the boundary effect is $\beta_0 = c(0)$. For a given temperature, the overlap-weighted effect $\beta_{\text{ov},\tau} = \mathbb{E}[\omega_\tau(H)c(H)]/\mathbb{E}[\omega_\tau(H)]$, with $\omega_\tau = e_\tau(1 - e_\tau)$ [9], is an intermediate target.

3 The result in brief

Take $\Delta(h) = h$, Gaussian outcomes with variance σ^2 , and mean functions bounded by B and L -Lipschitz; call this model $\mathcal{M}_{\text{Lip}}$. Let the logs come from one common temperature $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$, and let the operational gap vanish linearly at the boundary.

- (i) The Wald interval for the boundary effect is asymptotically valid uniformly over $\mathcal{M}_{\text{Lip}}$, and the exploration loss R_n tends to zero (Theorems 1 and 3).
- (ii) In $\mathcal{M}_{\text{Lip}}$, there exist two outcome laws with equal boundary effects and different population effects that differ only for the non-preferred action in a region where the operational gap is bounded below; no estimator is consistent for the population effect at both whenever $R_n \rightarrow 0$. Uniformly consistent estimation of the population effect over $\mathcal{M}_{\text{Lip}}$ requires divergent worst-case loss, for every design and estimator (Theorem 3).
- (iii) In the larger model of all bounded mean functions, uniform precision δ for the population effect requires worst-case loss of order at least $(\mathbb{E}\sqrt{g})^2/\delta^2$ as $\delta \rightarrow 0$, and this order does not depend on n (Theorem 2).

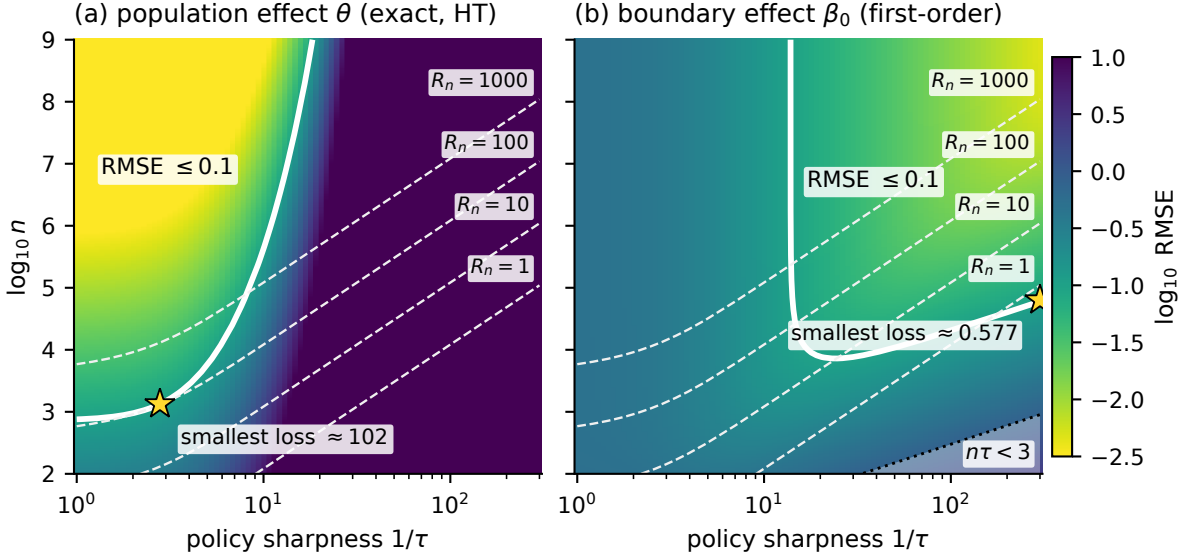


Figure 2: Root mean squared error ($\log_{10}$ scale) under one common temperature τ , for $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $c(h) = 1 + |h|$ and unit noise. (a) Exact root mean squared error of the Horvitz–Thompson estimator of θ , computed by numerical integration. (b) First-order approximation $\{(\beta_{\text{ov},\tau} - \beta_0)^2 + s_n^2/n\}^{1/2}$ to the root mean squared error of the estimator of Theorem 1, combining the overlap-target difference $\beta_{\text{ov},\tau} - \beta_0$ with the linearization variance s_n^2 . Monte Carlo checks with 4,000 replications, setting undefined estimates to zero as in Section 4, give ratios of Monte Carlo to approximate root mean squared error between 0.994 and 1.023 in the 16 checked settings with $n\tau \geq 3$, and 0.53 at $n\tau = 0.33$, where 0.25% of estimates were undefined; the shaded region $n\tau < 3$ marks this check’s range, not a general accuracy threshold. White curves enclose root mean squared error at most 0.1; dashed curves are levels of $R_n = n\mathbb{E}[g(H)q_\tau(H)]$; stars mark the smallest loss attaining the precision on the plotted grid ($1 \leq 1/\tau \leq 300$, $10^2 \leq n \leq 10^9$). Panel (a) concerns one estimator under one outcome law; lower bounds for all estimators are Theorems 2 and 3.

If instead every deviation carries a fixed cost, (i) fails as well (Proposition 1). Figure 2 shows the separation for the simulation design of Section 8. Under one common temperature the population effect reaches root mean squared error 0.1 only where the exploration loss is large, at least about 102 on the plotted grid, whereas the boundary effect reaches it with loss 0.58 at the edge of the grid and with less at larger n .

4 Boundary effects at vanishing exploration loss

Assumption 1 (Boundary model). $\Delta(h) = h$, $a^*(h) = \mathbf{1}\{h > 0\}$, and

(D1) H has a density $f \leq \bar{f}$, continuous at 0, with $f(0) > 0$;

(D2) $|\mu_a| \leq B$, and μ_a is L -Lipschitz on $[-u_0, u_0]$;

- (D3) $\sigma_a^2(h) = \text{Var}(Y \mid A = a, H = h)$ is continuous at 0 with $\sigma_1^2(0) + \sigma_0^2(0) > 0$, and $\mathbb{E}[Y^4 \mid A, H] \leq M_4$;
- (D4) $\mathbb{P}(A = 1 \mid H = h) = \text{sig}(h/\tau_n)$ with $\tau_n \rightarrow 0$, known;
- (D5) $g = \phi(|h|) \leq \bar{G}$ with $\phi(u) \leq \bar{\kappa}u$ on $[0, u_0]$; in the strong form $\phi'(0) = \kappa > 0$.

With $e_i = e_{\tau_n}(H_i)$, define

$$\begin{aligned} w_{1i} &= A_i(1 - e_i), & w_{0i} &= (1 - A_i)e_i, \\ \hat{D}_a &= \frac{1}{n} \sum_{i=1}^n w_{ai}, & \hat{m}_a &= \frac{\sum_{i=1}^n w_{ai} Y_i}{\sum_{i=1}^n w_{ai}}, & \hat{\beta} &= \hat{m}_1 - \hat{m}_0, \\ \hat{\psi}_i &= \frac{w_{1i}(Y_i - \hat{m}_1)}{\hat{D}_1} - \frac{w_{0i}(Y_i - \hat{m}_0)}{\hat{D}_0}, & \hat{s}^2 &= \frac{1}{n} \sum_{i=1}^n \hat{\psi}_i^2. \end{aligned} \quad (2)$$

$$(3)$$

If either weight sum is zero, set $\hat{\beta} = \hat{s}^2 = 0$ and report the interval $\mathbb{R}$; this event has probability tending to zero when $n\tau_n \rightarrow \infty$. Write $m_\tau = \mathbb{E}[\omega_\tau(H)]$, $\bar{\mu}_a = \mathbb{E}[\omega_\tau(H)\mu_a(H)]/m_\tau$,

$$\xi = \frac{w_1(Y - \bar{\mu}_1) - w_0(Y - \bar{\mu}_0)}{m_\tau}, \quad s_n^2 = \text{Var}(\xi),$$

so that $\bar{\mu}_1 - \bar{\mu}_0 = \beta_{\text{ov},\tau}$ and ξ is the linearization of $\hat{\beta} - \beta_{\text{ov},\tau}$.

Theorem 1 (Boundary inference). *Under Assumption 1, if $n\tau_n \rightarrow \infty$:*

- (i) $(\hat{\beta} - \beta_{\text{ov},\tau_n})/(s_n/\sqrt{n}) \Rightarrow N(0, 1)$, and $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau_n f(0)\}(1 + o(1))$;
- (ii) $|\beta_{\text{ov},\tau_n} - \beta_0| \leq C\tau_n$ for large n , for every $C > 4 \log 2 L \bar{f}/f(0)$;
- (iii) $\hat{s}^2/s_n^2 \rightarrow 1$ in probability;
- (iv) if also $n\tau_n^3 \rightarrow 0$, the Wald interval $\hat{\beta} \pm z_{1-\eta/2} \hat{s}/\sqrt{n}$ has asymptotic coverage $1 - \eta$ for β_0 ;
- (v) $R_n = n\mathbb{E}[g(H)q_{\tau_n}(H)] \leq n\tau_n^2 \bar{\kappa} \bar{f} \pi^2/6 + n\bar{G} \exp(-u_0/\tau_n)$, and in the strong form $R_n = n\tau_n^2 \kappa f(0) \pi^2/6 (1 + o(1))$.

In particular, for $\tau_n = n^{-\zeta}$ with $\zeta \in (1/2, 1)$, the interval in (iv) is asymptotically valid and $R_n \rightarrow 0$.

The three rate conditions are $n\tau_n \rightarrow \infty$ (the boundary sample grows, $\zeta < 1$), $n\tau_n^3 \rightarrow 0$ (bias is negligible relative to the standard error, $\zeta > 1/3$) and $n\tau_n^2 \rightarrow 0$ (the loss vanishes, $\zeta > 1/2$). The error is $|\hat{\beta} - \beta_0| = O_p((n\tau_n)^{-1/2} + \tau_n)$; we do not claim this rate is optimal. With $\zeta = 1/2$ the interval remains valid and the loss stays bounded. Differentiability of c at zero does not improve the bias order in general: a bounded effect equal to $1 + |h|^{3/2}$ near zero gives bias of order $\tau^{3/2}$. If c is $C^{1,1}$ and f is Lipschitz near zero, the bias is $O(\tau^2)$ and condition (iv) relaxes to $n\tau_n^5 \rightarrow 0$. Proofs are in Appendix A.

Vanishing loss does not mean that exploration disappears. The expected number of non-preferred actions is $n\mathbb{E}[q_{\tau_n}(H)] = 2 \log 2 f(0) n\tau_n(1 + o(1))$, which grows; the loss vanishes because these actions occur ever closer to the boundary, where the operational gap is small. The next result makes the role of the gap's shape explicit.

Proposition 1 (Shape of the operational gap). *Let (D1) and (D4) hold and $g = \phi(|h|) \leq \bar{G}$.*

- (a) *If $\phi(u) \leq \bar{\kappa}u^p$ on $[0, u_0]$ for some $p > 0$, then $R_n \leq 2\bar{\kappa}\bar{f}C_p n\tau_n^{p+1} + n\bar{G}\exp(-u_0/\tau_n)$ with $C_p = \int_0^\infty v^p \text{sig}(-v) dv < \infty$. Under the other conditions of Theorem 1, $\tau_n = n^{-\zeta}$ with $\max\{1/3, 1/(p+1)\} < \zeta < 1$ gives a valid interval for β_0 and $R_n \rightarrow 0$.*
- (b) *If $\phi(u) \geq \lambda_0 > 0$ for $u \in (0, u_0]$, then $R_n \geq 2 \log 2 \lambda_0 f(0) n\tau_n(1+o(1))$, which diverges whenever $n\tau_n \rightarrow \infty$, the condition under which Theorem 1 applies.*

Part (b) covers any cost incurred per deviation irrespective of context, such as a review or a payment. Under such costs the boundary effect is no cheaper to learn than the population effect, and the separation below does not hold in this form. The proof is in Appendix A.

5 Population effects require paid exploration

Fix $B > 0$ and $\sigma > 0$. Let $\mathcal{M}(B, \sigma)$ be the model in which $Y \mid A = a, H = h$ is $N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 are any measurable functions bounded by B . The model contains no structure that carries effects learned near the boundary to other contexts.

Theorem 2 (Information–exploration inequality). *Let $S_1, \dots, S_J$ be disjoint sets with $f_j = F(S_j) > 0$, each contained in $\{a^* = 0\}$ or in $\{a^* = 1\}$, and $g_j = \text{ess inf}_{S_j} g$. For every design, adaptive or not, and every estimator $\hat{\theta}$,*

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \cdot \left(\sup_{\mathcal{M}(B, \sigma)} R_n + \frac{\sigma^2 \pi^2}{B^2} \sum_j g_j \right) \geq \sigma^2 \left(\sum_j f_j \sqrt{g_j} \right)^2. \quad (4)$$

Consequently, if $F(g \geq \gamma) > 0$ for some $\gamma > 0$, uniform consistency requires $\sup R_n \rightarrow \infty$; and if $\sup \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2 \rightarrow 0$, then $\liminf_n \delta_n^2 \sup R_n \geq \sigma^2 (\mathbb{E} \sqrt{g(H)})^2$.

Neither side of (4) involves n : deployment scale does not substitute for exploration when the target is the population effect. At a fixed precision δ , (4) gives $\sup R_n \geq \sigma^2 (\sum_j f_j \sqrt{g_j})^2 / \delta^2 - \sigma^2 \pi^2 \sum_j g_j / B^2$, whose best value over partitions is at most $\sigma^2 (\mathbb{E} \sqrt{g})^2 / \delta^2$; the constant $(\mathbb{E} \sqrt{g})^2$ is reached only as $\delta \rightarrow 0$. If $\sum_j g_j > 0$, at $R_n = 0$ the bound is the finite floor $B^2 (\sum_j f_j \sqrt{g_j})^2 / (\pi^2 \sum_j g_j)$, consistent with the bounded error of a constant estimator.

Proposition 2 (Achievability). *Suppose $0 < \mathbb{E} \sqrt{g} < \infty$, $\delta_n > 0$ and $n\delta_n^2 \rightarrow \infty$. Write $C = \sigma^2 + B^2$. For $n\delta_n^2 > 4C$, define*

$$c_n = \frac{\mathbb{E} \sqrt{g}}{n\delta_n^2 / C - 4}, \quad q_n(h) = \begin{cases} \min\{1/2, c_n / \sqrt{g(h)}\}, & g(h) > 0, \\ 1/2, & g(h) = 0, \end{cases} \quad (5)$$

and let $e_n(h)$ be the corresponding probability of action 1. The Horvitz–Thompson estimator

$$\hat{\theta}_{\text{HT}} = \frac{1}{n} \sum_{i=1}^n \left\{ \frac{A_i Y_i}{e_n(H_i)} - \frac{(1 - A_i) Y_i}{1 - e_n(H_i)} \right\} \quad (6)$$

has $\sup_{\mathcal{M}(B,\sigma)} \mathbb{E}(\hat{\theta}_{\text{HT}} - \theta)^2 \leq \delta_n^2$ and $R_n \leq Cn(\mathbb{E}\sqrt{g})^2/(n\delta_n^2 - 4C) = C(\mathbb{E}\sqrt{g})^2\delta_n^{-2}(1 + o(1))$. If also $\mathbb{E}[g] < \infty$, uniform mixing with $q \equiv C/(n\delta_n^2 - 2C)$, for $n\delta_n^2 \geq 4C$, has the same uniform risk guarantee and $R_n = Cn\mathbb{E}[g]/(n\delta_n^2 - 2C) = C\mathbb{E}[g]\delta_n^{-2}(1 + o(1))$.

The order δ^{-2} and the functional $(\mathbb{E}\sqrt{g})^2$ appear in both the lower bound and the attainable loss, but the constants differ: the lower bound has σ^2 and the Horvitz–Thompson upper bound $\sigma^2 + B^2$, a factor that can be large when $B^2 \gg \sigma^2$. We do not establish the optimal constant. Proofs are in Appendix B.

6 The separation in one model

Theorem 2 is proved in $\mathcal{M}(B, \sigma)$, which is unrestricted away from the boundary. A lower bound over a large model does not transfer to a smaller one, so we state both sides in a model with global smoothness. Fix F satisfying (D1), $g = \phi(|h|)$ satisfying (D5), and $B, L, \sigma > 0$. Let $\mathcal{M}_{\text{Lip}}$ be the set of outcome laws with $Y \mid A = a, H = h \sim N(\mu_a(h), \sigma^2)$ and μ_0, μ_1 bounded by B and L -Lipschitz on $\mathbb{R}$. Every member of $\mathcal{M}_{\text{Lip}}$ satisfies (D2) and (D3).

Assumption 2. There are $0 < u_1 < u_2$ with $F([u_1, u_2]) > 0$ and $\gamma := \text{ess inf}_{h \in [u_1, u_2]} g(h) > 0$.

Theorem 3 (Separation). *Under Assumption 2, let $b(h) = \sin^2(\pi(h - u_1)/(u_2 - u_1))\mathbf{1}\{u_1 \leq h \leq u_2\}$, $t_{\max} = \min\{B, L(u_2 - u_1)/\pi\}$, and for $|t| \leq t_{\max}$ let μ^t have $\mu_1^t \equiv 0$ and $\mu_0^t = -tb$. Then $\mu^t \in \mathcal{M}_{\text{Lip}}$, all μ^t coincide on $(-\infty, u_1]$, $\beta_0(\mu^t) = 0$ and $\theta(\mu^t) = t\mathbb{E}[b(H)]$.*

(S1) *Under the common-temperature design $\tau_n = n^{-\zeta}$, $\zeta \in (1/2, 1)$, $R_n \rightarrow 0$ and $\sup_{\mu \in \mathcal{M}_{\text{Lip}}} |P_\mu\{\beta_0(\mu) \in \hat{\beta} \pm z_{1-\eta/2}\hat{s}/\sqrt{n}\} - (1 - \eta)| \rightarrow 0$.*

(S2) *For every design and every t , $\text{TV}(P_{\mu^0}^n, P_{\mu^t}^n) \leq R_n(\mu^0)/\gamma$. Hence if $R_n(\mu^0) \rightarrow 0$, for every $t_1 \in (0, t_{\max}]$ and every estimator, $\max_{t \in \{0, t_1\}} P_{\mu^t}(|\hat{\theta}_n - \theta(\mu^t)| \geq t_1\mathbb{E}[b]/2) \geq (1 - R_n(\mu^0)/\gamma)/2 \rightarrow 1/2$.*

(S3) *For every design and every estimator,*

$$\sup_{\mathcal{M}_{\text{Lip}}} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \frac{(\mathbb{E}[b])^2}{\sup_{\mathcal{M}_{\text{Lip}}} R_n/(\sigma^2\gamma) + \pi^2/t_{\max}^2}.$$

Both sides are now uniform over the same model: (S1) gives boundary intervals whose coverage converges uniformly, and (S3) shows that uniform consistency for the population effect requires unbounded worst-case loss. (S2) adds a statement at two fixed members. Because $\theta \in [-2B, 2B]$, uniform consistency in probability and in mean square are equivalent here after clipping an estimator to that interval. We do not claim that the constant of Theorem 2 holds in $\mathcal{M}_{\text{Lip}}$. If a submodel lets boundary data identify θ , the perturbation family is excluded and (S2)–(S3) do not apply. Proofs are in Appendix C.

7 Temperature as an exploration device

Theorem 4 (One common temperature). *Consider common-temperature designs with one deterministic τ_n per n . Assume $g = \phi(|\Delta|)$ with $\mathbb{E}[g] < \infty$; $|\Delta(H)|$ has a density at least $f_{\min} > 0$ on $[0, v_0]$; $\phi(u) \geq \underline{\kappa}u$ on $[0, v_0]$; and $D_{\max} = \text{ess sup } |\Delta(H)| < \infty$. Fix $\epsilon \in (0, D_{\max})$ and let $\nu_\epsilon = \max_j \mathbb{P}(|\Delta(H)| \geq D_{\max} - \epsilon, a^*(H) = j)$, which is positive. If an estimator has $\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \leq \delta_n^2$ with $\delta_n^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$ and $2n\delta_n^2 > \nu_\epsilon \sigma^2$, then*

$$R_n \geq \underline{\kappa} f_{\min} c_* n \min \left\{ v_0^2, \frac{(D_{\max} - \epsilon)^2}{\log^2 \{2n\delta_n^2 / (\nu_\epsilon \sigma^2)\}} \right\}, \quad c_* = \int_0^1 v \text{sig}(-v) dv.$$

This is a lower bound, and δ_n may be fixed. For fixed δ with $\delta^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$, the common-temperature loss is at least of order $n / \log^2 n$, whereas Proposition 2 attains the same uniform precision with loss bounded in n . The precision condition cannot be dropped: for $\delta \geq 2B$ the estimator that always outputs zero meets the target with no exploration. The theorem is stated in the unrestricted model $\mathcal{M}(B, \sigma)$; with extrapolation structure, such as a correctly specified parametric outcome model, a model-based estimator can do better. The reason for the cost is not that a common temperature explores in the wrong place, since the optimal design also explores most near the boundary; it is that the probability of the non-preferred action decays exponentially in $|\Delta|$, so observations far from the boundary become too rare unless the temperature is raised everywhere.

Remark 1 (Mixtures of temperatures). Theorem 4 does not extend to random mixtures of finite temperatures. With $|\Delta| = |H|$, $H \sim U(-1, 1)$ and $g = |H|$, let $r(u) = \text{sig}(-u)$ be the non-preferred probability at temperature 1, and mix temperature 1 with weight $w_n = \mathbb{E}[1/r(|H|)] / \{n\delta^2 / (\sigma^2 + B^2) - 2\}$ and temperature 1/ n otherwise. The Horvitz–Thompson estimator then has uniform mean squared error at most δ^2 over $\mathcal{M}(B, \sigma)$, and the loss is at most $(\sigma^2 + B^2) \mathbb{E}[1/r] \mathbb{E}[|H|r] / \delta^2 + O(1/n)$, bounded in n ; at $\delta = 0.1$ the factor $\mathbb{E}[1/r] \mathbb{E}[|H|r] / \delta^2$ equals 46.36. The lesson concerns a single common temperature, not temperature as such.

Proposition 3 (Context temperatures). *For $\Delta(h) \neq 0$ and $q(h) \in (0, 1/2)$, the context temperature $\tau(h) = |\Delta(h)| / \log\{(1 - q(h))/q(h)\}$ gives non-preferred probability $q(h)$; $q(h) = 1/2$ requires $\tau(h) = \infty$, and on $\{\Delta = 0\}$ every temperature gives $q = 1/2$. If $g = \phi(|\Delta|)$ with $\phi(0) = 0$, the design q_n of Proposition 2 is implementable with temperatures in $(0, \infty]$, and for each h with $\Delta(h) \neq 0$ and $g(h) > 0$, $\tau_n(h) \log(1/c_n) \rightarrow |\Delta(h)|$.*

The limit is pointwise. Where the cap binds, for example $0 < |h| \leq 4c_n^2$ when $g(h) = |h|$, the implementing temperature is $\tau = \infty$. When the gap is known only up to a multiplicative error, a Kantorovich bound (Lemma 1 in Appendix D) gives the worst-case cost of allocating by the misspecified gap relative to the correct one, in a limit of large inverse exploration moment. Proofs are in Appendix D.

8 Simulations

The simulations use $H \sim U(-1, 1)$, $\Delta(h) = h$, $g(h) = |h|$, $\mu_0(h) = h$, $c(h) = 1 + |h|$ (population effect 1.5, boundary effect 1) and $N(0, 1)$ noise. Code, seeds and result files

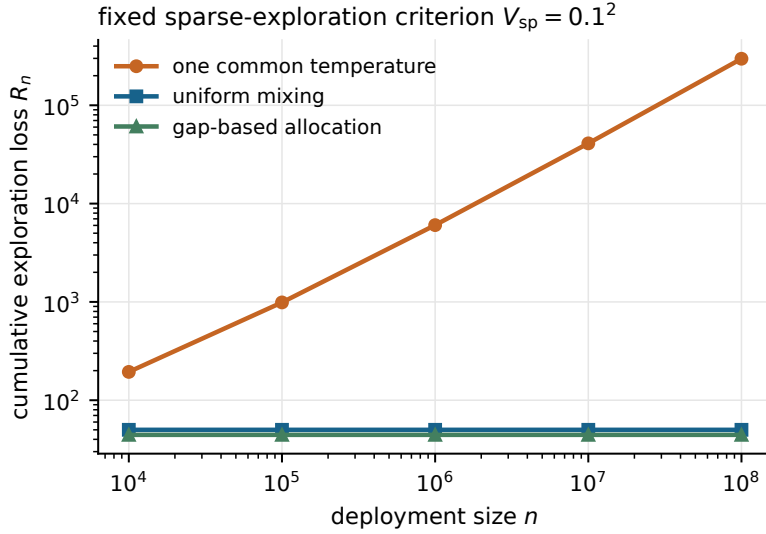


Figure 3: Cumulative exploration loss as deployment size increases, from closed-form calculations with the sparse-exploration criterion fixed at $V_{\text{sp}} = 0.1^2$. Designs are matched on this criterion, not on realized error; a context-dependent temperature gives the gap-based curve.

are in the repository cited in the data availability statement.

Exploration designs. For allocation comparisons we match the sparse-exploration criterion $V_{\text{sp}}(q) = \sigma^2 \mathbb{E}[1/q]/n$, an allocation criterion rather than the mean squared error of any estimator. At $n = 10^5$ and $V_{\text{sp}} = 0.01$, the closed-form exploration loss is 50.0 for uniform mixing, 44.4 for the gap-based design $q \propto g^{-1/2}$ (calibrated to V_{sp} and capped at $1/2$, rather than the c_n of Proposition 2), and 988.2 for a common temperature ($\tau = 0.110$), a ratio of 19.8 to uniform mixing (Figure 3; Figure 5 in Appendix E repeats it at $V_{\text{sp}} = 0.03^2$). Table 1 runs the augmented inverse-propensity estimator with a working model linear in $(1, H, |H|)$ separately in each arm, which is correctly specified for this design, on common random numbers across designs; the context-temperature implementation gives identical rows. At fixed V_{sp} the expected number of non-preferred actions stays bounded for uniform mixing and the gap-based design, so their nominal Wald coverage has no asymptotic justification as n grows; this does not apply to the common temperature. In a two-context example with masses $(0.5, 0.5)$, gaps $(0.2, 1)$, $\sigma = 1$, $B = 5$ and $\delta = 0.1$, the lower bound (4) evaluates to 51.9 against the asymptotic constant 52.4, and the common-temperature loss has local growth exponent 0.808 against the predicted $1 - \Delta_{\min}/\Delta_{\max} = 0.8$.

Boundary inference. Table 2 reports the estimator of Theorem 1. Exploration loss matches the constant in Theorem 1(v). With $\zeta = 0.6$ coverage is close to nominal; at $n = 10^6$, 5,000 replications give 0.9458 (MCSE 0.0032), with the estimated standard error at 0.993 of the Monte Carlo standard deviation. With a very small boundary sample the interval under-covers: 0.887 at $n\tau_n \approx 10$.

δ	design	RMSE	coverage	MCSE	explored	loss R_n
0.10	uniform mixing	0.098	0.950	0.011	100	49.9
0.10	$q \propto g^{-1/2}$ (= context temp.)	0.103	0.932	0.013	134	44.5
0.10	common temperature	0.097	0.965	0.009	7607	988.5
0.05	uniform mixing	0.052	0.955	0.010	400	200.3
0.05	$q \propto g^{-1/2}$ (= context temp.)	0.053	0.940	0.012	532	178.0
0.05	common temperature	0.050	0.948	0.011	9191	1440.1

Table 1: Exploration designs at $n = 10^5$, 400 replications with common random numbers. Coverage is of the population effect by Wald intervals; MCSE uses the observed coverage. “Explored” is the mean number of non-preferred actions.

ζ	n	$n\tau_n$	reps	RMSE	coverage of β_0	MCSE	loss R_n (Thm. 1)
0.60	10,000	39.8	1,000	0.215	0.949	0.007	1.3×10^{-1} (1.3×10^{-1})
0.60	100,000	100.0	500	0.141	0.944	0.010	8.2×10^{-2} (8.2×10^{-2})
0.60	1,000,000	251.2	200	0.082	0.970	0.012	5.1×10^{-2} (5.2×10^{-2})
0.75	10,000	10.0	1,000	0.464	0.887	0.010	8.1×10^{-3} (8.2×10^{-3})
0.75	100,000	17.8	500	0.339	0.924	0.012	2.6×10^{-3} (2.6×10^{-3})
0.75	1,000,000	31.6	200	0.247	0.930	0.018	8.2×10^{-4} (8.2×10^{-4})
0.60	1,000,000	251.2	5,000	0.090	0.946	0.003	–

Table 2: Boundary inference under common temperature $\tau_n = n^{-\zeta}$. Coverage is of $\beta_0 = c(0)$. Loss is the Monte Carlo mean with the Theorem 1(v) constant in parentheses. The last row is a separate 5,000-replication run.

Temperature sweep. Figure 4 fixes $n = 100,000$ and sharpens a common temperature, with 1,000 replications per temperature. Wald coverage of the population effect falls from 0.949 at $1/\tau = 1$ to 0.085 (Horvitz–Thompson) and 0.031 (augmented estimator with the fitted working model) at $1/\tau = 120$. Coverage of the boundary effect rises from 0.000 to 0.943: at $1/\tau = 1$ the overlap-target difference $\beta_{\text{ov},\tau} - \beta_0$ is 0.48, so the interval is centred at the wrong target, and the difference shrinks as τ falls. The exploration loss falls from 17,056 to 5.7. Boundary coverage improves over this range because $n\tau$ stays at least 833; at fixed n , lowering the temperature further drives $n\tau$ toward zero and the boundary interval then fails as well (Table 2). For the Horvitz–Thompson estimator the Monte Carlo error understates the exact standard deviation once the policy is sharp: at $1/\tau = 120$ its Monte Carlo root mean squared error is 0.36, while its exact standard deviation is of order 10^{22} , carried by rare, very large terms that the replications do not observe (Figure 6 in Appendix E).

A regression-discontinuity benchmark. The estimator of Theorem 1 uses the known assignment probabilities and only the τ -neighbourhood of the boundary. A fuzzy local-linear regression-discontinuity estimator at $\Delta = 0$ ignores the probabilities and uses a bandwidth chosen by the analyst. Table 3 in Appendix E compares the two at $n = 10^5$

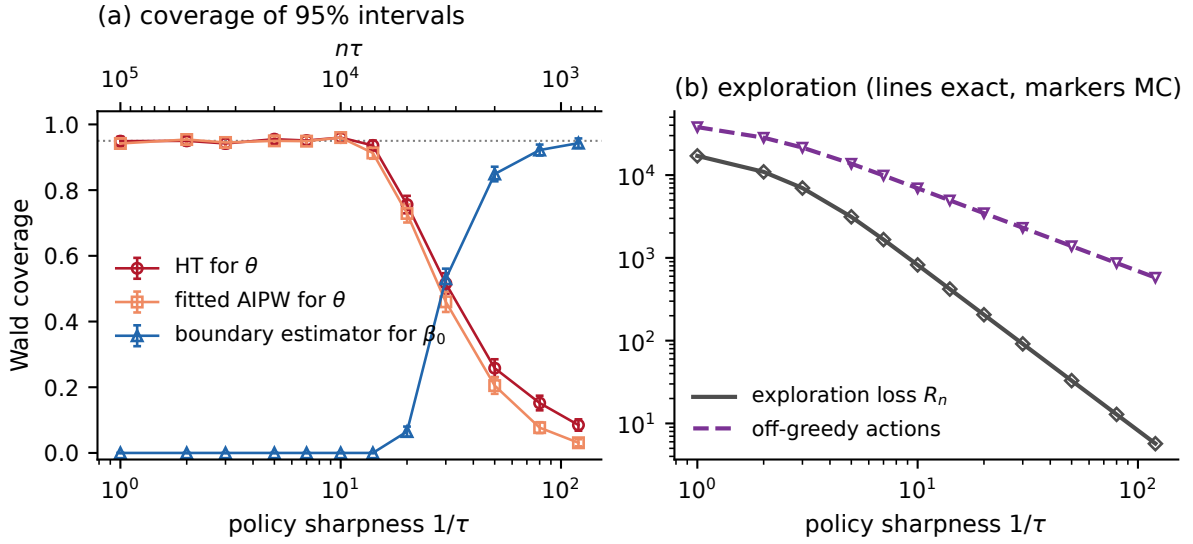


Figure 4: Common temperature at fixed $n = 100,000$, 1,000 replications per temperature, same design as Figure 2. (a) Wald coverage with 95% Monte Carlo intervals for the population effect (Horvitz–Thompson and augmented estimator with a fitted working model) and for the boundary effect; the top axis shows $n\tau$, which decreases to the right. (b) Cumulative exploration loss and expected number of non-preferred actions; lines are exact expectations, markers Monte Carlo.

on the same logs. At $1/\tau \leq 30$ neither is reliable: the estimator of Theorem 1 is centred at $\beta_{\text{ov},\tau}$ rather than β_0 , and the discontinuity estimator has a weak first stage within the bandwidth. At $1/\tau = 120$ both cover at close to the nominal level; with the wider bandwidth the discontinuity estimator has the smaller error there, and at $1/\tau = 1000$ it has the smaller error at both bandwidths, helped by mean functions that are linear on each side of the boundary, which removes its bias at wide bandwidths. The regression-discontinuity estimator is therefore a competitive and often more efficient way to analyse such logs when the policy is sharp; the separation concerns which target is learnable and at what exploration cost, and does not depend on which boundary estimator is used.

9 Discussion

The results separate two uses of logs from sharpening policies. If the question is how outcomes respond where the deployed policy is indifferent, and the operational cost vanishes at the boundary, a growing deployment permits increasingly precise local inference, uniformly over a Lipschitz model, while cumulative exploration loss vanishes; the number of exploratory actions still grows, but their operational cost falls. If the question is the population average effect, uniformly consistent estimation requires divergent worst-case exploration loss, and deployment size does not remove that requirement. Reporting an effect estimated from such logs therefore requires saying which of the two quantities it is.

For design, the relevant choice is not whether to use temperature but how. With one temperature shared by all contexts, the probability of the non-preferred action decays exponentially in the policy’s confidence, so reaching confident contexts forces a temperature that is costly near the boundary; uniform mixing, context-dependent temperatures or temperature mixtures with a vanishing weight on a moderate temperature avoid this. Allocating by the gap saves further loss when gaps are heterogeneous; in the closed-form costs of Section 8 the saving relative to uniform mixing is about eleven percent.

Limitations. The policy is known and assignments are logged; the score and preferred action do not learn from outcomes; the setting is a single decision per unit with independent contexts; the boundary results are one-dimensional; the lower bounds assume Gaussian outcomes with known variance and exclude models in which boundary data identify the population effect; the separation requires an operational cost that vanishes at the boundary; and the treatment is strategy assignment under a fixed generator, not the realized text. Extending the results to learned scores, multidimensional scores (for example the conditional effect given $\Delta(H) = 0$) and multi-turn conversations is left open.

Data availability statement

No new participant data were collected or analysed. The results are theoretical, deterministic numerical illustrations, or simulations. The simulation code, the result files (with arguments, random seeds and code commits), plain-text versions of the proofs, and the scripts that generate every number, table and figure in the paper are available in the repository

<https://github.com/kunhailP/exploration-cost-boundary-population>

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The author states no conflict of interest.

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The conducted research is not related to either human or animals use.

Use of AI-assisted tools

[TO CONFIRM against the publisher’s policy] Generative AI tools were used to assist in drafting text, checking proofs, and writing simulation code. The author reviewed all content and takes full responsibility for it.

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A Proofs for Section 4

Write $e = e_\tau(H) = \text{sig}(H/\tau) = \mathbb{P}(A = 1 \mid H)$ and $\omega = e(1 - e) = \Omega(H/\tau)$ with $\Omega(u) = \exp(u)/\{1 + \exp(u)\}^2$, and $m_\tau = \mathbb{E}\omega$. We use

$$\begin{aligned} \int \Omega = 1, \quad \int |u|\Omega(u) du = 2 \log 2, \quad \int \Omega(u)\text{sig}(u) du = \tfrac{1}{2}, \\ \Omega(u) \leq \exp(-|u|), \quad \int_0^\infty \text{sig}(-v) dv = \log 2, \quad \int_0^\infty v \text{sig}(-v) dv = \tfrac{\pi^2}{12}. \end{aligned}$$

A.1 Proof of Theorem 1

Step 1 (kernel integrals). For ϱ bounded and continuous at 0, $j \geq 1$ and $r, s \geq 0$, substituting $h = \tau u$ and using dominated convergence with dominating function $\|\varrho\|_\infty \bar{f}\Omega(u)$,

$$\mathbb{E}[\Omega(H/\tau)^j (1 - e)^r e^s \varrho(H)] = \tau f(0) \varrho(0) \int \Omega^j \text{sig}(-u)^r \text{sig}(u)^s du + o(\tau).$$

In particular $m_\tau = \tau f(0)(1 + o(1))$.

Step 2 (bias). $\beta_{\text{ov},\tau} - \beta_0 = \mathbb{E}[\omega\{c(H) - c(0)\}]/m_\tau$. On $|H| \leq u_0$, $|c(h) - c(0)| \leq 2L|h|$ and $\mathbb{E}[\omega|H|] \leq \bar{f}\tau^2 \int |u|\Omega = 2 \log 2 \bar{f}\tau^2$. On $|H| > u_0$, $\omega \leq \exp(-u_0/\tau)$ and $|c(h) - c(0)| \leq 4B$. Hence $|\beta_{\text{ov},\tau} - \beta_0| \leq (4 \log 2 L \bar{f}\tau^2 + 4B \exp(-u_0/\tau))/m_\tau = (4 \log 2 L \bar{f}/f(0))\tau(1 + o(1))$.

Step 3 (linearization). Since $\mathbb{E}[w_1 | H] = \omega$ and $\mathbb{E}[w_1 Y | H] = \omega \mu_1$, $\mathbb{E}[w_1(Y - \bar{\mu}_1)] = 0$, and likewise for arm 0. Define $T_a = n^{-1} \sum_i w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$, so that $\hat{m}_a - \bar{\mu}_a = (m_\tau/\hat{D}_a)T_a$, $T_1 - T_0 = n^{-1} \sum_i \xi_i$ and $\mathbb{E}\xi = 0$. Because $w_1 w_0 = 0$ and $\mathbb{E}[A(1-e)^2 | H] = \omega(1-e)$,

$$s_n^2 = m_\tau^{-2} \mathbb{E}[\omega\{(1-e)(\sigma_1^2 + (\mu_1 - \bar{\mu}_1)^2) + e(\sigma_0^2 + (\mu_0 - \bar{\mu}_0)^2)\}]. \quad (7)$$

By Step 1, $\bar{\mu}_a \rightarrow \mu_a(0)$; the terms in $(\mu_a - \bar{\mu}_a)^2$ are $o(\tau)$ by dominated convergence; and the variance terms equal $\tau f(0)\sigma_a^2(0)/2 + o(\tau)$. Hence $s_n^2 = \{\sigma_1^2(0) + \sigma_0^2(0)\}/\{2\tau f(0)\}(1 + o(1))$.

Step 4 (central limit theorem). Since $0 \leq w_a \leq 1$, $w_a^4 \leq w_a$, and $\mathbb{E}[(Y - \bar{\mu}_a)^4 | A, H] \leq C(M_4, B)$, we get $\mathbb{E}[w_a^4(Y - \bar{\mu}_a)^4] \leq C m_\tau$ and $\mathbb{E}\xi^4 \leq C' m_\tau^{-3}$. The Lyapunov ratio $n^{-1}\mathbb{E}\xi^4/s_n^4 = O((n\tau)^{-1}) \rightarrow 0$, so $n^{-1/2} \sum \xi_i/s_n \Rightarrow N(0, 1)$. Also $\text{Var}(\hat{D}_a)/m_\tau^2 \leq 1/(nm_\tau) \rightarrow 0$, so $\hat{D}_a/m_\tau \rightarrow 1$ in probability and $m_\tau/\hat{D}_a - 1 = O_p((n\tau)^{-1/2})$. Now

$$\hat{\beta} - \beta_{\text{ov}, \tau} = (T_1 - T_0) + (m_\tau/\hat{D}_1 - 1)T_1 - (m_\tau/\hat{D}_0 - 1)T_0, \quad (8)$$

with $T_a = O_p((n\tau)^{-1/2})$. The remainder is $O_p((n\tau)^{-1}) = o_p(s_n/\sqrt{n})$, and Slutsky's lemma gives (i). The event that a weight sum is zero has probability at most $2(1 - m_\tau)^n \leq 2 \exp(-nm_\tau) \rightarrow 0$, because $\mathbb{P}(w_a > 0) \geq \mathbb{E}w_a = m_\tau$.

Step 5 (variance estimator). Write $\xi_{ai} = w_{ai}(Y_i - \bar{\mu}_a)/m_\tau$. First, $n^{-1} \sum \xi_i^2/s_n^2$ has mean one and variance at most $\mathbb{E}\xi^4/(ns_n^4) = O((n\tau)^{-1})$. Second,

$$\hat{\psi}_{1i} - \xi_{1i} = w_{1i}(Y_i - \bar{\mu}_1)(1/\hat{D}_1 - 1/m_\tau) + w_{1i}(\bar{\mu}_1 - \hat{m}_1)/\hat{D}_1.$$

The mean square of the first term is $(n^{-1} \sum \xi_{1i}^2)(m_\tau/\hat{D}_1 - 1)^2 = O_p(s_n^2)O_p((n\tau)^{-1})$. For the second, $w_1^2 \leq w_1$ gives mean square at most $(\hat{m}_1 - \bar{\mu}_1)^2/\hat{D}_1$; by Step 4 and $\hat{D}_1 = m_\tau(1 + o_p(1))$ this is $O_p(1/(nm_\tau^2))$, which is $O_p((n\tau)^{-1})$ relative to $s_n^2 \asymp \tau^{-1}$. Arm 0 is identical. By Minkowski's and the Cauchy-Schwarz inequalities, $|n^{-1} \sum \hat{\psi}_i^2 - n^{-1} \sum \xi_i^2| = o_p(s_n^2)$, which gives (iii).

Step 6 (centering at β_0). $(\beta_{\text{ov}, \tau} - \beta_0)/(s_n/\sqrt{n}) = O(\tau)O((n\tau)^{1/2}) = O((n\tau^3)^{1/2})$. With (i), (iii) and Slutsky's lemma this gives (iv).

Step 7 (exploration loss). The non-preferred probability is $q_\tau(h) = \text{sig}(-|h|/\tau)$. On $|h| \leq u_0$, $n\mathbb{E}[\phi(|H|)\text{sig}(-|H|/\tau)] \leq n\bar{\kappa}\bar{f} \cdot 2\tau^2 \int_0^\infty v \text{sig}(-v)dv = n\bar{\kappa}\bar{f}\tau^2\pi^2/6$, and the rest is at most $n\bar{G}\exp(-u_0/\tau)$. In the strong form, on $v \leq u_0/\tau$, $\phi(\tau v)/\tau \rightarrow \kappa v$ and $f(\tau v) \rightarrow f(0)$ with dominating function $\bar{\kappa}v\bar{f}\text{sig}(-v)$, which gives the asymptotic equality. For $\tau_n = n^{-\zeta}$: $n\tau_n \rightarrow \infty$ iff $\zeta < 1$; $n\tau_n^3 \rightarrow 0$ iff $\zeta > 1/3$; $n\tau_n^2 \rightarrow 0$ iff $\zeta > 1/2$; and $n\exp(-u_0n^\zeta) \rightarrow 0$. The expected number of non-preferred actions is $n\mathbb{E}[q_\tau(H)] = n\tau \int f(\tau u)\text{sig}(-|u|)du = 2\log 2 f(0) n\tau(1 + o(1))$ by the same dominated convergence argument. $\square$

Smoother effects. If $|c(h) - c(0) - c'(0)h| \leq M_2 h^2$ and f is L_f -Lipschitz near 0, then $\mathbb{E}[\omega H] = \tau^2 \int u\Omega(u)\{f(\tau u) - f(0)\}du = O(\tau^3)$ up to an exponentially small tail, and the quadratic remainder contributes at most $M_2 \bar{f}\tau^3 \int u^2 \Omega/m_\tau$; the bias is $O(\tau^2)$. Differentiability of c alone is not enough: a bounded effect equal to $c(h) = 1 + |h|^{3/2}$ near zero gives bias $\tau^{3/2} \int |u|^{3/2} \Omega(u)du (1 + o(1))$.

A.2 Proof of Proposition 1

(a) Substituting $h = \tau v$ on $|h| \leq u_0$ and using $\text{sig}(-v) \leq \exp(-v)$ elsewhere,

$$R_n \leq n\bar{\kappa}\bar{f} \int_{|h| \leq u_0} |h|^p \text{sig}(-|h|/\tau) dh + n\bar{G} \exp(-u_0/\tau) \leq 2\bar{\kappa}\bar{f}C_p n\tau^{p+1} + n\bar{G} \exp(-u_0/\tau).$$

Here $C_p = \Gamma(p+1)(1-2^{-p})\zeta_R(p+1)$ with ζ_R the Riemann zeta function, which follows from expanding $\text{sig}(-v) = \sum_{k \geq 1} (-1)^{k+1} \exp(-kv)$; for example $C_{1/2} \approx 0.678$, $C_1 = \pi^2/12$ and $C_2 \approx 1.803$. Validity of the interval uses only (D1)–(D4) and $1/3 < \zeta < 1$, by Theorem 1(iv), and $n\tau_n^{p+1} \rightarrow 0$ iff $\zeta > 1/(p+1)$.

(b) Restricting to $0 < |H| \leq u_0$ and substituting $h = \tau u$,

$$R_n \geq n\lambda_0\tau \int_{|u| \leq u_0/\tau} f(\tau u) \text{sig}(-|u|) du = 2 \log 2 \lambda_0 f(0) n\tau(1 + o(1)),$$

by dominated convergence with dominating function $\bar{f} \text{sig}(-|u|)$. $\square$

B Proofs for Section 5

Proof of Theorem 2. *Submodel.* For $\vartheta \in [-B, B]^J$ set $\mu_{a^*(h)}^\vartheta(h) = 0$, and on S_j set $\mu_{\bar{a}(h)}^\vartheta(h) = s_j \vartheta_j$ with $s_j = +1$ if $\bar{a} = 1$ on S_j and $s_j = -1$ otherwise; elsewhere $\mu_{\bar{a}(h)}^\vartheta(h) = 0$. Then $\mu^\vartheta \in \mathcal{M}(B, \sigma)$ and $\theta(\mu^\vartheta) = \sum_j f_j \vartheta_j$.

Information. The sample density is $\prod_i dF(H_i) \pi_i(A_i | H_i, \mathcal{F}_{i-1}) \varphi_\sigma(Y_i - \mu_{A_i}^\vartheta(H_i))$, where the assignment kernels π_i are known functions of observed data and do not depend on ϑ . The score for ϑ_j is $\sum_i \mathbf{1}\{H_i \in S_j, A_i = \bar{a}(H_i)\} s_j (Y_i - s_j \vartheta_j) / \sigma^2$. Its summands are martingale differences, so the Fisher information matrix is diagonal with $I_j(\vartheta) = \mathbb{E}_\vartheta N_j / \sigma^2$, where $N_j = \sum_i \mathbf{1}\{H_i \in S_j, A_i = \bar{a}(H_i)\}$.

van Trees. Take the product prior with marginal density $B^{-1} \cos^2(\pi x / (2B))$ on $[-B, B]$, whose Fisher information is $J_0 = \pi^2 / B^2$. With $\bar{N}_j = \int \mathbb{E}_\vartheta N_j \lambda(d\vartheta)$, the multi-variate van Trees inequality [43], maximized over the direction vector, gives

$$\sup_{\mathcal{M}(B, \sigma)} \mathbb{E}(\hat{\theta} - \theta)^2 \geq \sum_j \frac{f_j^2}{\bar{N}_j / \sigma^2 + J_0} = \sum_j \frac{f_j \sigma^2}{a_j}, \quad a_j = \frac{\bar{N}_j + \sigma^2 J_0}{f_j}.$$

Cauchy-Schwarz and loss. Writing $\sigma \sum_j f_j \sqrt{g_j} = \sum_j \sqrt{f_j g_j a_j} \cdot \sigma \sqrt{f_j / a_j}$,

$$\sigma^2 \left(\sum_j f_j \sqrt{g_j} \right)^2 \leq \left(\sum_j f_j g_j a_j \right) \left(\sum_j f_j \sigma^2 / a_j \right).$$

Here $\sum_j f_j g_j a_j = \sum_j g_j \bar{N}_j + \sigma^2 J_0 \sum_j g_j$, and since $g \geq g_j$ almost surely on S_j , $\sum_j g_j N_j \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\sum_j g_j \bar{N}_j \leq \sup R_n$. This proves (4).

Consequences. With $J = 1$ and $S_1 = \{g \geq \gamma\} \cap \{a^* = j\}$ of positive mass for some j , (4) gives $\sup R_n \geq g_1 \{\sigma^2 f_1^2 / \sup \mathbb{E}(\hat{\theta} - \theta)^2 - \sigma^2 \pi^2 / B^2\} \rightarrow \infty$. For the constant, fix a partition into level sets of g intersected with the two preferred-action sides; (4) gives $\liminf \delta_n^2 \sup R_n \geq \sigma^2 (\sum f_j \sqrt{g_j})^2$, and the lower sums $\sum F(S) \text{ess inf}_S \sqrt{g}$ increase to $\mathbb{E} \sqrt{g}$ as the level mesh shrinks. $\square$

Proof of Proposition 2. The Horvitz–Thompson estimator is unbiased with variance at most $n^{-1}C\mathbb{E}[1/e + 1/(1-e)] = n^{-1}C\mathbb{E}[1/q + 1/(1-q)] \leq n^{-1}C(\mathbb{E}[1/q] + 2)$, since $\{e, 1-e\} = \{q, 1-q\}$ and $q \leq 1/2$. Pointwise $1/q = \max\{2, \sqrt{g}/c_n\} \leq \sqrt{g}/c_n + 2$, and the choice of c_n gives $\mathbb{E}[1/q] + 2 \leq n\delta_n^2/C$. The loss is $n\mathbb{E}[gq] \leq nc_n\mathbb{E}\sqrt{g}$. For uniform mixing with $q = C/(n\delta_n^2 - 2C) \leq 1/2$, the same variance bound gives $C(1/q + 2)/n = \delta_n^2$, and the loss is $nq\mathbb{E}[g] = C\mathbb{E}[g]n/(n\delta_n^2 - 2C)$. Without the correction, $q = C/(n\delta_n^2)$ gives risk $C/\{nq(1-q)\} > \delta_n^2$ at $\mu_0 = \mu_1 \equiv B$, where the bound is attained. $\square$

C Proof of Theorem 3

The function b satisfies $0 \leq b \leq 1$, is $\pi/(u_2 - u_1)$ -Lipschitz on $\mathbb{R}$, vanishes outside $[u_1, u_2]$, and has $\mathbb{E}b(H) > 0$ because H has a density and $F([u_1, u_2]) > 0$. Hence $|tb| \leq B$ and tb is L -Lipschitz for $|t| \leq t_{\max}$, so $\mu^t \in \mathcal{M}_{\text{Lip}}$; $c^t = tb$, $\beta_0(\mu^t) = 0$, and $\theta(\mu^t) = t\mathbb{E}b(H)$. Every observation affected by the perturbation has $A = 0$ and $H \in [u_1, u_2] \subset (0, \infty)$, where $a^* = 1$, so it is a non-preferred action. Let $N = \sum_i \mathbf{1}\{A_i = 0, H_i \in [u_1, u_2]\}$.

(S1): uniform boundary coverage. The loss statement is Theorem 1(v), since the design does not depend on μ . For coverage we show that the error terms in the proof of Theorem 1 are bounded by constants that do not depend on $\mu \in \mathcal{M}_{\text{Lip}}$. Here $\sigma_a^2 \equiv \sigma^2$, and m_τ , $\hat{D}_a$ and the event of a zero weight sum depend only on (H_i, A_i) , whose law does not depend on μ . Throughout, C denotes a constant depending only on $(B, L, \sigma, \bar{f})$.

Variance bounds. Since $(1-e) + e = 1$ and $|\mu_a - \bar{\mu}_a| \leq 2B$, (7) gives

$$\sigma^2/m_\tau \leq s_n^2 \leq (\sigma^2 + 4B^2)/m_\tau.$$

Berry–Esseen. Given (A, H) , $Y - \bar{\mu}_a$ is Gaussian with variance σ^2 and mean bounded by $2B$, so $\mathbb{E}[|Y - \bar{\mu}_a|^3 \mid A, H] \leq C$ and, as $w_a^3 \leq w_a$ and $\mathbb{E}w_a = m_\tau$, $\mathbb{E}|\xi|^3 \leq Cm_\tau^{-2}$. The Berry–Esseen inequality for independent and identically distributed summands gives

$$\sup_x \left| \mathbb{P}_\mu \left(\frac{\sum_i \xi_i}{\sqrt{ns_n}} \leq x \right) - \Phi(x) \right| \leq \frac{C\mathbb{E}|\xi|^3}{\sqrt{n}s_n^3} \leq \frac{C}{\sigma^3\sqrt{nm_\tau}},$$

uniformly in μ .

Remainder. In (8), $\text{Var}(T_a) \leq \mathbb{E}[w_a^2(Y - \bar{\mu}_a)^2]/(nm_\tau^2) \leq (\sigma^2 + 4B^2)/(nm_\tau)$, and $\text{Var}(\hat{D}_a/m_\tau) \leq 1/(nm_\tau)$ does not depend on μ . By Chebyshev’s inequality, for every $\varepsilon > 0$ there is M with $\sup_\mu \mathbb{P}_\mu(|T_a| > M(nm_\tau)^{-1/2}) < \varepsilon$ and $\mathbb{P}(|m_\tau/\hat{D}_a - 1| > M(nm_\tau)^{-1/2}) < \varepsilon$ for large n . Dividing by $s_n/\sqrt{n} \geq \sigma(nm_\tau)^{-1/2}$, the remainder in units of $s_n/\sqrt{n}$ is $O_p((nm_\tau)^{-1/2})$ uniformly in μ .

Variance estimator. Each bound in Step 5 of the proof of Theorem 1 involves only $\mathbb{E}\xi^4 \leq Cm_\tau^{-3}$, $s_n^2 \geq \sigma^2/m_\tau$, the bounds on T_a above and the law of $\hat{D}_a$; hence $\sup_\mu \mathbb{P}_\mu(|\hat{s}^2/s_n^2 - 1| > \varepsilon) \rightarrow 0$ for every $\varepsilon > 0$.

Bias. Global Lipschitz continuity gives, with no tail term, $|\beta_{\text{ov},\tau} - \beta_0| \leq 2L\mathbb{E}[\omega|H|]/m_\tau \leq 4\log 2L\bar{f}\tau^2/m_\tau$, so in units of $s_n/\sqrt{n}$ the bias is at most $C\tau^2\sqrt{n/m_\tau} = O((n\tau^3)^{1/2})$ uniformly in μ , using $m_\tau = \tau f(0)(1 + o(1))$.

Coverage. Write $Z_n = (\hat{\beta} - \beta_{\text{ov},\tau})\sqrt{n}/s_n$ and $b_n = (\beta_{\text{ov},\tau} - \beta_0)\sqrt{n}/s_n$. The interval covers β_0 iff $|Z_n + b_n| \leq z_{1-\eta/2}\hat{s}/s_n$. By the three displays above, for every $\varepsilon > 0$ there are events of $\mathbb{P}_\mu$ -probability at least $1-\varepsilon$ for all μ and large n , on which $|Z_n - n^{-1/2} \sum_i \xi_i/s_n| \leq \varepsilon$, $|b_n| \leq \varepsilon$ and $|\hat{s}/s_n - 1| \leq \varepsilon$. Hence the coverage probability lies within $\varepsilon + C/\sqrt{nm\tau} + \sup_{|x| \leq z+3\varepsilon} \varphi(x) \cdot 6\varepsilon$ of $1-\eta$ for all $\mu \in \mathcal{M}_{\text{Lip}}$, where φ is the standard normal density, and letting $\varepsilon \downarrow 0$ proves (S1).

(S2). The sample densities under μ^0 and μ^t coincide on $E = \{N = 0\}$, so $P_0(E) = P_t(E)$, and for any event A , $|P_0(A) - P_t(A)| = |P_0(A \cap E^c) - P_t(A \cap E^c)| \leq \max\{P_0(E^c), P_t(E^c)\} = P_0(E^c)$. Thus $\text{TV} \leq P_0(N \geq 1) \leq \mathbb{E}_0 N$. Because $g \geq \gamma$ F -almost surely on $[u_1, u_2]$ and $H_i \sim F$, $\gamma N \leq \sum_i g(H_i) \mathbf{1}\{A_i = \bar{a}(H_i)\}$ almost surely, so $\mathbb{E}_0 N \leq R_n(\mu^0)/\gamma$. Le Cam's two-point argument with separation $t_1 \mathbb{E} b$ gives the displayed bound.

(S3). The score of $t \mapsto \mu^t$ is $\sum_i \mathbf{1}\{A_i = 0\} b(H_i) \{-(Y_i + t b(H_i))\}/\sigma^2$, with information $I(t) = \mathbb{E}_t \sum_i \mathbf{1}\{A_i = 0\} b(H_i)^2/\sigma^2 \leq \mathbb{E}_t N/\sigma^2 \leq R_n(\mu^t)/(\gamma\sigma^2)$. With the prior $t_{\max}^{-1} \cos^2(\pi t/(2t_{\max}))$ on $[-t_{\max}, t_{\max}]$, whose information is $\pi^2/t_{\max}^2$, the van Trees inequality for $\psi(t) = t \mathbb{E} b$ gives the bound. $\square$

D Proofs and a lemma for Section 7

Proof of Theorem 4. Take the single cell $S = \{|\Delta| \geq D_{\max} - \epsilon, a^* = j\}$, with j attaining ν_ϵ . There $q_\tau \leq \exp(-(D_{\max} - \epsilon)/\tau)$, so $\bar{N} \leq n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau)$, and the van Trees step of Theorem 2 with $J = 1$ gives $\delta_n^2 \geq \nu_\epsilon^2 \sigma^2 / (n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau) + \sigma^2 \pi^2 / B^2)$. Once $\delta_n^2 \leq \nu_\epsilon^2 B^2 / (2\pi^2)$, this forces $n\nu_\epsilon \exp(-(D_{\max} - \epsilon)/\tau) \geq \nu_\epsilon^2 \sigma^2 / (2\delta_n^2)$, that is, $\tau_n \geq (D_{\max} - \epsilon) / \log\{2n\delta_n^2 / (\nu_\epsilon \sigma^2)\}$, where the logarithm is positive because $2n\delta_n^2 > \nu_\epsilon \sigma^2$. Nothing here requires $\delta_n \rightarrow 0$. On the other hand

$$R_n = n\mathbb{E}[\phi(|\Delta|)\text{sig}(-|\Delta|/\tau)] \geq n\underline{\kappa} f_{\min} \tau^2 \int_0^{v_0/\tau} v \text{sig}(-v) dv \geq n\underline{\kappa} f_{\min} c_* \min\{\tau, v_0\}^2.$$

Combining the two bounds proves the claim. With finitely many score values, the same argument gives $\tau \geq \Delta_{\max} / \log(C'n\delta^2)$ for δ small enough, and the $\Delta_{\min}$ cell, with mass f , contributes at least $n f \phi(\Delta_{\min}) \exp(-\Delta_{\min}/\tau)/2$, which is of order $n^{1-\Delta_{\min}/\Delta_{\max}}$ up to logarithmic factors. $\square$

Remark 1. Write $u = |H|$, $C = \sigma^2 + B^2$, $r(u) = \text{sig}(-u)$, and $q_n = w_n r + (1 - w_n) \text{sig}(-nu) \leq 1/2$ for the mixture. Since $q_n \geq w_n r$, $\mathbb{E}[1/q_n] \leq \mathbb{E}[1/r]/w_n = n\delta^2/C - 2$, and the variance bound in the proof of Proposition 2 gives uniform risk at most $C(\mathbb{E}[1/q_n] + 2)/n \leq \delta^2$. The loss is $n\mathbb{E}[uq_n] \leq nw_n \mathbb{E}[ur] + n\mathbb{E}[u \text{sig}(-nu)] = C\mathbb{E}[1/r]\mathbb{E}[ur]n/(n\delta^2 - 2C) + O(1/n)$, which equals $C\mathbb{E}[1/r]\mathbb{E}[ur]/\delta^2 + O(1/n)$.

Proof of Proposition 3. Solving $\text{sig}(-|\Delta|/\tau) = q$ gives the formula. If $g(h) = 0$, then $q_n(h) = 1/2$, which is implementable by $\tau = \infty$ (and by any τ when $\Delta(h) = 0$);

if $\Delta(h) \neq 0$ and the cap binds, $q_n(h) = 1/2$ again requires $\tau = \infty$. If $\Delta(h) \neq 0$ and $g(h) > 0$, then $q_n(h) = c_n/\sqrt{g(h)} < 1/2$ once $c_n < \sqrt{g(h)}/2$, and $\log\{(1 - q_n)/q_n\} = \log(1/c_n) + \frac{1}{2} \log g(h) + \log(1 - q_n)$ with $c_n \rightarrow 0$, which gives the limit. $\square$

Misspecified gaps. Suppose the allocation uses $\hat{g}$ in place of g . Designs are compared at a common value of the sparse-exploration criterion $\sigma^2 \mathbb{E}[1/q]/n$, which is an allocation criterion, not the mean squared error of an estimator. Let $\Lambda(\rho) = (1 + \sqrt{\rho})^2/(4\sqrt{\rho})$.

Lemma 1. *Assume $0 < \mathbb{E}\sqrt{g} < \infty$ and $0 < a \leq \hat{g}/g \leq b < \infty$ on $\{g > 0\}$, with $\hat{g} = 0$ on $\{g = 0\}$, and let $\rho = b/a$.*

- (a) *If $F(g > 0) = 1$ and the uncapped designs $q = c/\sqrt{\hat{g}}$ and $q' = c'/\sqrt{g}$, calibrated to the same criterion, are at most $1/2$ almost surely, the ratio of their losses is at most $\Lambda(\rho)$. The bound is attained when the measure $w \propto \sqrt{g} dF$ has a set of mass exactly one half.*
- (b) *For the capped designs $q_c = \min\{1/2, c/\sqrt{\hat{g}}\}$ and $q'_c = \min\{1/2, c'/\sqrt{g}\}$, equal to $1/2$ where the gap is zero and calibrated to a common value $T = \mathbb{E}[1/q_c] = \mathbb{E}[1/q'_c]$, the loss ratio converges, as $T \rightarrow \infty$, to $\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/(\mathbb{E}\sqrt{g})^2 \leq \Lambda(\rho)$.*

If $\text{ess inf } g = 0$, the uncapped design exceeds $1/2$ on a set of positive mass for every $c > 0$, so part (a) is vacuous and part (b) is the relevant statement; this includes $g(h) = |h|$. A common rescaling of $\hat{g}$ changes nothing. Gap-based allocation beats uniform mixing for every admissible $\hat{g}$ if $\mathbb{E}[g]/(\mathbb{E}\sqrt{g})^2 > \Lambda(\rho)$.

Proof of Lemma 1. (a) With $q = c/\sqrt{\hat{g}}$ and $\mathbb{E}[1/q] = T$, $c = \mathbb{E}\sqrt{\hat{g}}/T$ and the loss is $n\mathbb{E}[\sqrt{\hat{g}}]\mathbb{E}[g/\sqrt{\hat{g}}]/T$, against $n(\mathbb{E}\sqrt{g})^2/T$ for $\hat{g} = g$. With $s = (\hat{g}/g)^{1/2} \in [\sqrt{a}, \sqrt{b}]$ and the probability measure $w \propto \sqrt{g} dF$, the ratio is $\mathbb{E}_w[s]\mathbb{E}_w[1/s]$, which Kantorovich's inequality bounds by $(\sqrt{a} + \sqrt{b})^2/(4\sqrt{ab}) = \Lambda(b/a)$. If w has a set of mass one half, setting $s = \sqrt{a}$ there and $s = \sqrt{b}$ elsewhere attains the bound. With atoms the bound need not be attained: for $F = (0.99, 0.01)$, $g = (1, 100)$ and $\rho = 36$, uniform mixing costs 1.675 times the optimum, below $\Lambda(36) = 2.042$, yet every admissible $\hat{g}$ costs at most 1.347 times the optimum.

(b) By dominated convergence, $c\mathbb{E}[1/q_c] = \mathbb{E}\max\{2c, \sqrt{\hat{g}}\} \rightarrow \mathbb{E}\sqrt{\hat{g}}$ and $\mathbb{E}[gq_c]/c \rightarrow \mathbb{E}[g/\sqrt{\hat{g}}]$, with $g/\sqrt{\hat{g}} = 0$ where $g = 0$, as $c \rightarrow 0$; likewise for q' . Calibration to a common $T \rightarrow \infty$ forces $c, c' \rightarrow 0$ with $cT \rightarrow \mathbb{E}\sqrt{\hat{g}}$ and $c'T \rightarrow \mathbb{E}\sqrt{g}$, so the loss ratio $\mathbb{E}[gq_c]/\mathbb{E}[gq'_c]$ converges to the stated limit, which is bounded as in (a). $\square$

E Additional figures and tables

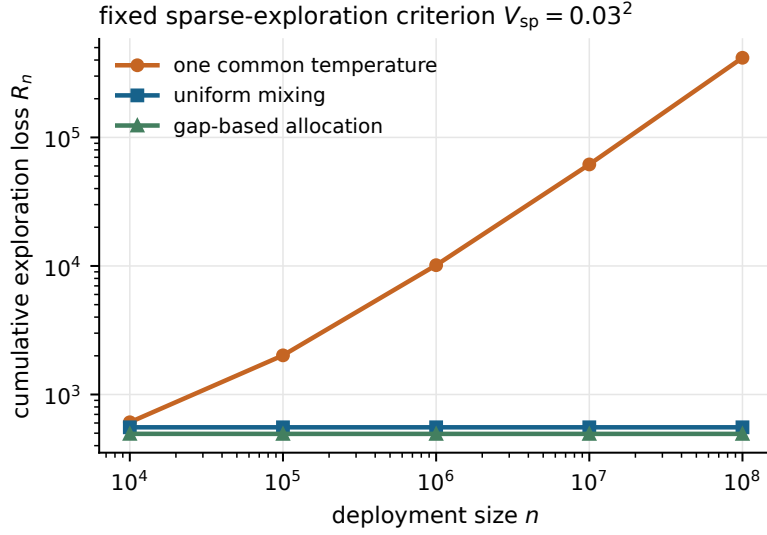


Figure 5: As Figure 3, with the sparse-exploration criterion fixed at $V_{\text{sp}} = 0.03^2$.

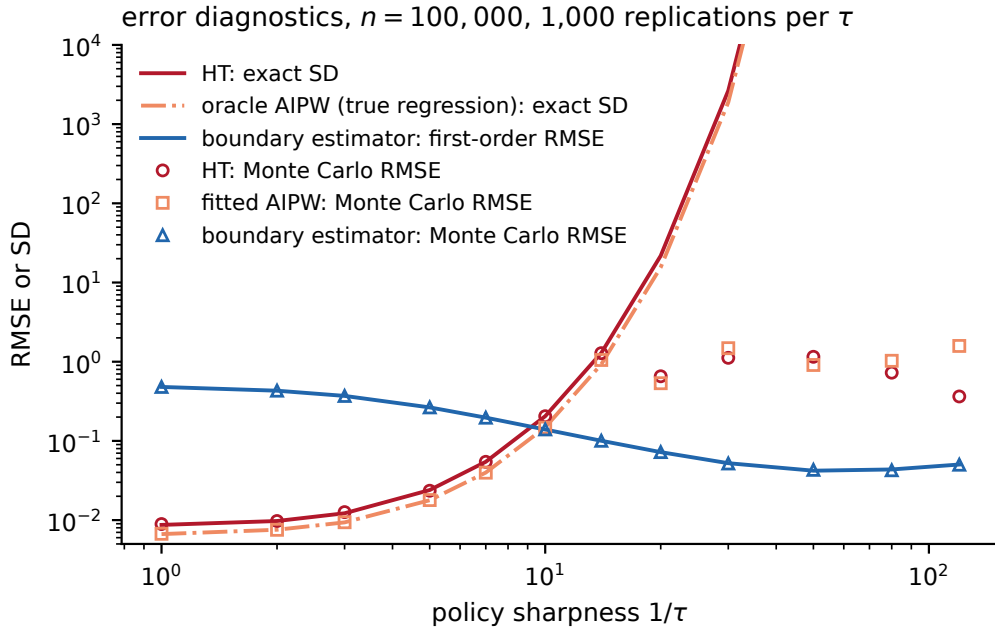


Figure 6: Error diagnostics for the runs of Figure 4. Lines: exact standard deviation of the Horvitz–Thompson estimator and of an oracle augmented estimator that uses the true regression (a reference, not an estimator used in the simulation), both for θ , and the first-order approximation to the root mean squared error of the boundary estimator of β_0 . Markers: Monte Carlo root mean squared error of the Horvitz–Thompson estimator, the augmented estimator with the fitted working model, and the boundary estimator.

$1/\tau$	b	Theorem 1 estimator		local-linear RD		$n\tau$
		RMSE	coverage	RMSE	coverage	
10	0.05	0.139	0.000	164.8	1.000	10,000
10	0.20	0.140	0.000	34.6	0.996	10,000
30	0.05	0.052	0.536	252.6	1.000	3,333
30	0.20	0.051	0.545	0.153	0.915	3,333
120	0.05	0.050	0.945	0.261	0.962	833
120	0.20	0.050	0.947	0.044	0.940	833
1000	0.05	0.141	0.951	0.073	0.953	100
1000	0.20	0.138	0.956	0.032	0.957	100

Table 3: Boundary effect at $n = 10^5$, 1,000 replications: estimator of Theorem 1 (known probabilities) and a fuzzy local-linear regression-discontinuity estimator at $\Delta = 0$ with a triangular kernel, bandwidth b and heteroskedasticity-robust delta-method standard errors. The design and the exploration loss are the same for both. Very large errors of the discontinuity estimator at larger τ reflect a weak first stage within the bandwidth.